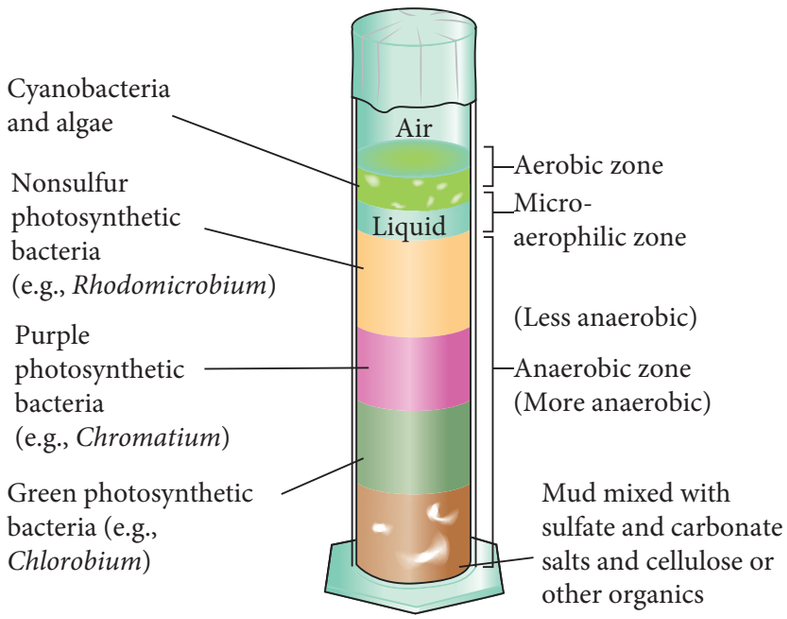


Figure 10.2: Winogradsky column

Sulphur cycle. Therefore, he is considered as the '**Father of Soil Microbiology**'.

M. W. Beijerinck (1888) isolated root nodule bacteria in pure culture from nodules in legumes and named them as *Bacillus radicola*. Thus, he is considered as the '**Father of Microbial ecology**'.

Beijerinck and Winogradsky (1890) developed the enrichment culture technique for isolation of soil organisms, proved independently that transformation of nitrogen in nature is largely due to the activities of various groups of soil microorganisms (1891). Therefore, they are considered as '**Pioneers in Soil Bacteriology**'.

10.3 Components of Soil

The soil is composed of five major components

- Mineral matter
- Water
- Air
- Organic matter
- Living organisms

One gram of soil contains approximately one million microorganisms. The soil has organic matter, soil solution and soil air. All these components are affected by the activities of microorganisms. Soil is a constantly changing medium. The soil solution in agricultural soil has ions like K^+, Na^{++}, Mg^{++}, Ca^{++}, Fe^+, S^-, NO_3, SO_4, PO_4 and others.

These ions are very essential in culture media. In a fertile soil, these elements in mineral form are supplemented by organic compound, derived from the decomposition of animal and plant residues. Thus the soil is an excellent natural medium for growth of microorganisms.

10.4 Soil Microorganisms

Soil contain five major groups of microorganisms. They are Bacteria, Actinomycetes, Fungi, Algae and Protozoa (Table 10.1).

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